

Project ID:

25-26J-198

1. Topic (12 words max)

Civil Legal Document Summarization in Sri Lanka

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	25-26J-198
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

As of today, the legal domain provides a large volume of documents including wills, contracts, business documents and legal opinions. These documents are complex, highly formal, specific, and harder to understand. In this situation, it takes lot of time to extract relevant information quickly. Legal industry becomes increasingly digitalize, the challenge of understanding and discovery has become more critical than ever.

One of the main problems is that methods used in document review are traditional manual methods. It is not scalable. Lawyers, judges, and researchers spend lot of time reading these lengthy documents to find outcomes relevant to their cases. Especially they may also make various errors while reading in a limited amount of time or documents is large.

Automatic legal summarization aims to answer this problem by using natural language processing (NLP) and machine learning (ML) techniques to generate correct summarize outcomes. But this will face some problems. In legal documents includes complex sentences and specialized vocabulary. Furthermore, it is essential to protect the legal authenticity of those documents, as misinterpretation can have a major impact.

In summary, the core problem in legal summarization is creating documents are trustworthy, can work large amount of data sets and provide correct outcomes. These systems must also understand the complex language used in legal documents.

References:

1. M. Koniari et al., Evaluation of Automatic Legal Text Summarization Techniques for Greek Judgements, Information, MDPI, 2023.
2. S. D. Kavila et al., "An Automatic Legal Document Summarization and Search Using Hybrid System," Advances in Intelligent Systems and Computing, vol. 199, pp. 229–236, 2012.
3. D. Anand and R. Wagh, "Effective Deep Learning Approaches for Summarization of Legal Texts," Journal of King Saud University – Computer and Information Sciences, vol. 34, no. 5, pp. 2141–2150, 2022.
4. S. D. Kavila, V. Puli, G. S. V. P. Raju, and R. Bandaru, "An Automatic Legal Document Summarization and Search Using Hybrid System," Advances in Intelligent Systems and Computing, vol. 199, pp. 229–236, 2013.
5. M. Akter et al., "Advancements in Natural Language Processing: Implications for Legal Summarization," International Journal of Innovative Trends in Information Technology, Apr. 2024.
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7. D. Jain, M. D. Borah, and A. Biswas, "Summarization of Legal Documents: Where Are We Now and the Way Forward," Computer Science Review, vol. 40, Art. 100388, 2021.
8. S. Liu et al., "Low-Resource Court Judgment Summarization for Common Law Systems," Information Processing & Management, vol. 61, Art. 103796, 2024.
9. T. Turan and E. U. Küçüksille, "Summarization, Prediction, and Analysis of Turkish Constitutional Court Decisions with Explainable Artificial Intelligence and a Hybrid NLP Method," IEEE Access, 2025.
10. W. S. El-Kassas et al., "Multi-Task Deep Learning for Legal Document Translation, Summarization and Multi-Label Classification," Proceedings of the AICCC Conference, ACM, 2018.
11. S. Sharma and P. P. Singh, "Advancing Legal Document Summarization: Introducing an Approach Using a Recursive Summarization Algorithm," SN Computer Science, Springer, 2024.
12. X. Wang and Y. C. Wu, Empowering Legal Justice with AI: A Reinforcement Learning SAC-VAE Framework for Advanced Legal Text Summarization, PLOS ONE, 2024.
13. Y. Sun, F. Yang, X. Wang, and H. Dong, Automatic Generation of Draft Procuratorial Suggestions Based on an Extractive Summarization Method: BERTSLCA, Mathematical Problems in Engineering, Hindawi (Wiley), 2021.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed system is a legal document smart analysis platform that seeks to help individuals better understand and analyze legal documents. It helps both professionals and non-experts through the use of artificial intelligence to simplify complicated legal content.

Automatic summarization is one of the system's core elements, which distills lengthy legal documents into brief, readable summaries of important facts, issues, and conclusions. This allows users to grasp the case's key points instantly without being required to read the entire document.

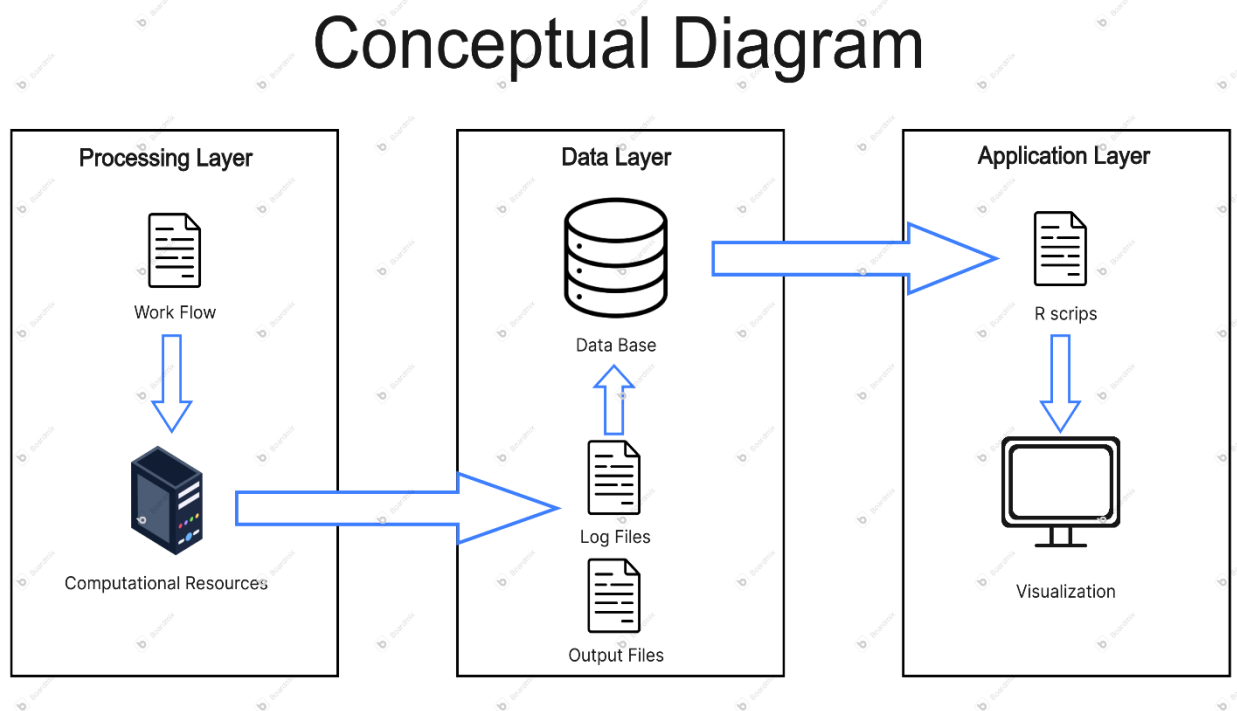
The system includes a chatbot that allows users to navigate the document using natural language. The user may ask questions, ask for explanations, or seek information on a particular section. This makes usability better and provides a more friendly experience.

Argument scoring is another important aspect that analyzes the quality and coherence of legal arguments presented in the document. It provides users with a sense of the quality of legal reasoning used in the case.

The platform also includes an interactive dashboard with a timeline generator. This tool takes significant procedural events like filings, hearings, and judgments and puts them in chronological order. This graphical timeline enables the user to get an immediate feel for the case flow.

All these four features, summarization, chatbot interaction, argument scoring, and the timeline dashboard, form a full solution for legal research, learning, and well-informed decision-making.

Conceptual Diagram:



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Legal expertise

Researchers must collaborate with legal experts who work with the law to understand case judgments and government rules. This helps in training tools and models to produce accurate system outputs. They also identify key terms, clauses, and obligations in documents, ensuring no critical information is missed.

Natural Language Processing (NLP) & Machine Learning (ML)

The system uses Natural Language Processing (NLP) and Machine Learning (ML) to train and understand legal documents. It summarizes complex legal texts by identifying key details and converting them into clear, brief summaries, keeping the original meaning. Features like summarization, chatbot, and argument analysis are powered by ML models such as BERT and Legal-BERT, fine-tuned for the unique language and structure of legal texts to ensure accurate, reliable results.

Conversational AI and System Integration

To build the chat feature, developers need to understand how chatbots work and how to handle natural language questions because the system must understand what the user is asking and respond correctly using the information from legal documents. Without this knowledge, the chatbot wouldn't be able to interpret questions or give accurate answers from the uploaded files.

User Experience and Interface Design

Designing an easy-to-use interface for users like legal professionals, law students, and the public is important. The interface should show summaries, chatbot answers, and argument analysis clearly and simply, making complex legal information easy to understand and interact with.

Data Requirements and Annotation

Having good and useful legal data is very important to making this project successful. For example, public court judgments, case law databases, and legal texts with notes or labels. To train the system well, may also need special labels for things like argument strength, important people or terms, and summary quality. This might involve working with legal experts or law students to add those labels correctly.

8. Objectives and Novelty
Main Objective:

The main objective is automatically generating short and clear summaries of complex legal documents such as judgements and contracts. It identifies key parts like facts, legal issues, decisions then generate summary using NLP techniques. The summarization function helps reduce the time and effort required to review lengthy legal texts. And this will help lawyers, students and the public to quickly understand the main message of the case.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Jayawardhana M C IT22361868	Legal Summarizer	First upload or enter a legal document (judgment, contract) Text preprocessing: Remove footnotes, headlines irrelevant symbols and split text into sentences and paragraphs. Identify the key factors. Use models to find the most important sentences. Can use models like BERT. Use pre-trained models to rewrite content into a shorter. Can use BART or Legal BERT. After, improve grammar and readability. Finally, Display the output.	This system designed for the legal domain. It identifies and summarizes all legal documents such as judge's decisions and arguments. It uses special models for trained data. Not normal texts. It will create short and clear summaries. And the system helps users to quickly understand the complex legal documents.

		Optionally users can adjust the output and download or read it.	It will be useful to lawyers, students and any other people who need legal information.
P.R Reid IT22575944	Chat Assistance	Identify common user queries and define the chatbot's scope, focusing on questions related to case facts, legal terms, arguments, and decisions. Design and implement a clean, user-friendly chat interface that integrates seamlessly with the main application. Apply natural language processing to interpret user input, detect intent, and map queries to the appropriate content within the legal document. Integrate the chatbot with both the summarization output and structured document data to ensure accurate and relevant responses.	The chatbot enhances the system by introducing an interactive, conversational interface that allows users to query legal documents using natural language. Unlike traditional static tools, it provides context aware responses and direct document navigation, making complex legal content more accessible and easier to explore.

		Implement response generation using pre-trained language models capable of producing context-aware, legally accurate answers. Enable navigation functionality that allows the chatbot to highlight or redirect users to specific sections of the document based on their queries. Conduct testing using legal documents, evaluate the accuracy and clarity of responses, and refine the chatbot based on user feedback.	
Manoli G D C IT22352958	Legal Timeline Generator	Define the types of legal events to extract (e.g., case filing, hearings, bail decisions, appeals, judgments, etc.) based on standard legal procedural flows. Build a system to read and clean legal documents from files like PDFs or Word documents, or plain text.	The Legal Timeline Generator simplifies complex legal documents by automatically extracting and presenting key case events (such as filing dates, hearings, bail orders, and final decisions) in chronological order.

		Apply Natural Language Processing (NLP) techniques to detect date mentions, event related sentences, and legal action verbs. Use Named Entity Recognition (NER), keyword matching, or a machine learning based classifier to identify and classify relevant legal events. Arrange all detected events in chronological order by checking and comparing their associated dates. Create a user-friendly visual timeline interface (e.g., vertical/horizontal scroll timeline) that displays date-tagged legal events. Enable expandable timeline events that, when clicked, reveal context-specific snippets from the document. Evaluate the system using diverse real-world legal documents and validate event detection accuracy, date extraction and timeline clarity.	This feature is especially helpful for users who need a quick and clear overview of a case's procedural history without having to read through lengthy documents. It is particularly useful for students, lawyers, and researchers who want to analyze or compare legal case timelines. This kind of visual and interactive timeline tool is rarely available in existing legal analysis platforms, making it a valuable and innovative component of the system.
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DAYARATHNE R.D.T.N IT22578396	Argument Strength Scorer	Set rules for scoring legal arguments based on how well the argument uses legal sources and laws, how clearly the arguments are written, how relevant they are to the case, and how much they influenced the final decision. Develop a system to recognize and extract arguments from both sides in legal documents. Apply argument mining techniques (find key points, reasons, and evidence) and semantic analysis (looks at the meaning and clarity of the words) to assess argument structure and strength. Use machine learning models to score arguments based on defined criteria and legal impact. Design an easy-to-understand interface that shows the argument scores clearly, so users can quickly compare which side made stronger points.	The Argument Strength Scorer looks closely at the arguments in a legal case and measures how strong each one is. It compares the arguments from both sides to see which is better. It uses things like how many legal sources are used, how clearly the argument is written, and how much the argument helped decide the case. This is helpful for students and new lawyers because it teaches them what makes an argument strong, something most other legal tools don't show.
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Jayawardhana M C IT22361868	This project applies core information technology concepts to solve real world legal problems using techniques. There are NLP to analyze these documents and machine learning to summarize main parts in legal documents. There is also a user interface. This will be a collection of AI, data science and software engineering, making it highly relevant to IT specialization.
P.R Reid IT22575944	The Chatbot Assistance component matches the Information Technology specialization as it makes use of fundamental IT abilities like language processing, interface design, and system integration. the Chatbot Assistance component aligns with the Information Technology specialization. By allowing users to ask questions and receive precise, understandable answers, it facilitates user interaction with legal documents.
Manoli G D C IT22352958	This is applies core Information Technology concepts to solve real-world legal problems by developing a Legal Timeline Generator. It uses Natural Language Processing (NLP) to extract important events and dates from legal documents and machine learning to classify the types of events. The extracted information is presented in a user-friendly visual timeline, making complex legal case flows easier to understand. This component combines AI, data science, and software engineering, making it highly relevant to the IT specialization.
DAYARATHNE R.D.T.N IT22578396	The Argument Strength Scorer uses key Information Technology concepts like natural language processing (NLP), machine learning, and text analysis to understand and rate legal arguments. It looks at things like how arguments are written, which sources they cite, and whether they helped win the case. This helps make legal analysis easier using smart computer tools.

10. Supervisor's details

	Title	First Name	Last Name	Signature
Supervisor	Ms	Lochana	Rajamanthri	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.